

Module 4 : Applications of Quantum Mechanics

(Lecture Notes for Engineering Physics : BAPHY105)

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Disclaimer

These lecture notes do not claim originality. They are an assimilation of existing materials from various standard textbooks and lecture notes, compiled and reorganized to suit the structure and requirements of the course syllabus.

Note

These lecture notes are continuously updated based on classroom discussions and questions asked by students. There may still be mistakes, and if you find any, please feel free to write to me at vishnudath.kn@vit.ac.in.

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Module 4

Applications of Quantum Mechanics

Introduction

In the previous module we introduced the fundamental postulates of quantum mechanics, including the Schrödinger equation, operators, and the probabilistic interpretation of the wavefunction. In this chapter we apply those principles to a number of important physical systems. These examples illustrate how quantum mechanics predicts phenomena that have no classical counterpart.

We begin with the **infinite square well**, one of the simplest and most important exactly solvable models in quantum mechanics. Despite its simplicity, this system clearly demonstrates several fundamental quantum features such as the quantization of energy, the existence of stationary states, and the role of boundary conditions in determining the allowed wavefunctions.

We then extend the discussion to the **three-dimensional infinite potential well**, where the method of separation of variables allows us to construct solutions from the one-dimensional case. This example also introduces the concept of **degeneracy**, where different quantum states can share the same energy.

Finally, we discuss **quantum tunnelling**, a purely quantum-mechanical phenomenon in which a particle can penetrate and pass through a potential barrier even when its energy is less than the barrier height. Tunnelling plays an important role in many physical processes and technological applications, including α -decay in nuclear physics and the operation of devices such as the scanning tunnelling microscope (STM).

Through these examples, we see how the abstract formalism of quantum mechanics leads to concrete and experimentally observable predictions.

4.1 The Infinite Square Well

4.1.1 The Potential

The infinite square well is defined as

$$V(x) = \begin{cases} 0, & 0 \leq x \leq a, \\ \infty, & \text{otherwise.} \end{cases} \quad (4.1)$$

A particle is thus confined within the region $[0, a]$ and cannot exist outside. See Fig. 4.1.

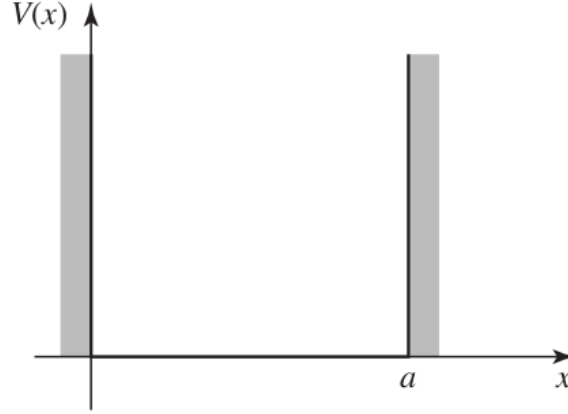


Figure 4.1: The infinite square well of width a . (Adapted from *Introduction to Quantum Mechanics* by David J. Griffiths and Darelle F. Schroeter, 3rd edition, for instructional use only.)

4.1.2 Time-Independent Schrödinger Equation

Inside the well ($0 < x < a$), the potential is zero, so the time-independent Schrödinger equation reads

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} = E\psi.$$

Rearranging,

$$\frac{d^2\psi}{dx^2} = -k^2\psi, \quad k \equiv \frac{\sqrt{2mE}}{\hbar}.$$

This is the simple harmonic oscillator equation, whose general solution is

$$\psi(x) = A \sin(kx) + B \cos(kx).$$

4.1.3 Boundary Conditions

Since the wavefunction must vanish where $V(x) = \infty$:

$$\psi(0) = 0, \quad \psi(a) = 0.$$

At $x = 0$, we find $\psi(0) = B = 0$. Hence

$$\psi(x) = A \sin(kx).$$

At $x = a$, the condition $\psi(a) = 0$ requires

$$A \sin(ka) = 0.$$

For a nontrivial solution ($A \neq 0$), this implies

$$ka = n\pi, \quad n = 1, 2, 3, \dots$$

so¹

$$k_n = \frac{n\pi}{a}.$$

¹The case $n = 0$ corresponds to $k = 0$, which would give $\psi(x) = 0$ everywhere, the trivial non-physical solution. Negative n values lead to $\sin(-n\pi x/a) = -\sin(n\pi x/a)$, which differ only by an overall sign and hence do not represent new physical states.

4.1.4 Energy Eigenvalues

The corresponding energies are

$$E_n = \frac{\hbar^2 k_n^2}{2m} = \frac{n^2 \pi^2 \hbar^2}{2ma^2}, \quad n = 1, 2, 3, \dots$$

Thus the spectrum is discrete and quadratic in n .

4.1.5 Normalized Wavefunctions

Normalization requires

$$\int_0^a |\psi(x)|^2 dx = 1.$$

Substituting $\psi(x) = A \sin(n\pi x/a)$,

$$|A|^2 \int_0^a \sin^2\left(\frac{n\pi x}{a}\right) dx = 1.$$

Using $\int_0^a \sin^2\left(\frac{n\pi x}{a}\right) dx = \frac{a}{2}$, we find

$$|A|^2 \frac{a}{2} = 1 \quad \Rightarrow \quad A = \sqrt{\frac{2}{a}}.$$

Hence the normalized eigenfunctions are

$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right), \quad n = 1, 2, 3, \dots$$

The lowest state, $\psi_1(x)$, is called the **ground state**, with energy

$$E_1 = \frac{\pi^2 \hbar^2}{2ma^2}.$$

The next state, $\psi_2(x)$, is the **first excited state**, with energy

$$E_2 = \frac{4\pi^2 \hbar^2}{2ma^2},$$

and $\psi_3(x)$ is the **second excited state**, with energy

$$E_3 = \frac{9\pi^2 \hbar^2}{2ma^2}.$$

In general, the n th eigenfunction corresponds to the $(n - 1)$ th excited state.

The first three normalized wavefunctions $\psi_1(x)$, $\psi_2(x)$, $\psi_3(x)$ are illustrated in Fig. 4.2, which shows their sinusoidal form within the infinite square well.

4.1.6 Properties of Eigenfunctions

The normalized eigenfunctions of the infinite square well are

$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right), \quad n = 1, 2, 3, \dots$$

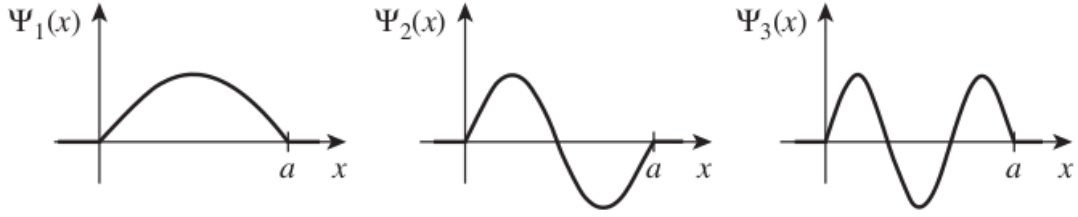


Figure 4.2: The first three stationary states $\psi_1(x)$, $\psi_2(x)$, and $\psi_3(x)$ for the infinite square well of width a . (Adapted from *Introduction to Quantum Mechanics* by David J. Griffiths and Darrell F. Schroeter, 3rd edition, for instructional use only.)

Proof of Orthonormality

Consider the inner product of two eigenfunctions:

$$I_{mn} = \int_0^a \psi_m(x) \psi_n(x) dx = \frac{2}{a} \int_0^a \sin\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi x}{a}\right) dx.$$

Using the trigonometric identity

$$\sin \alpha \sin \beta = \frac{1}{2} [\cos(\alpha - \beta) - \cos(\alpha + \beta)],$$

we obtain

$$I_{mn} = \frac{1}{a} \int_0^a \left\{ \cos\left(\frac{(m-n)\pi x}{a}\right) - \cos\left(\frac{(m+n)\pi x}{a}\right) \right\} dx.$$

Evaluating the integrals:

$$\int_0^a \cos\left(\frac{(m \pm n)\pi x}{a}\right) dx = \frac{a}{(m \pm n)\pi} \sin\left(\frac{(m \pm n)\pi x}{a}\right) \Big|_0^a.$$

At the limits, $\sin[(m \pm n)\pi] = 0$, so both terms vanish whenever $m \neq n$. Thus,

$$I_{mn} = 0 \quad \text{for } m \neq n.$$

For $m = n$, we have

$$I_{nn} = \frac{2}{a} \int_0^a \sin^2\left(\frac{n\pi x}{a}\right) dx = \frac{2}{a} \cdot \frac{a}{2} = 1.$$

Hence,

$$\int_0^a \psi_m(x) \psi_n(x) dx = \delta_{mn},$$

showing that the eigenfunctions form an orthonormal set.

Completeness and Probabilistic Interpretation

The eigenfunctions also form a complete basis. Any square-integrable wavefunction $f(x)$ defined on $[0, a]$ can be expanded as

$$f(x) = \sum_{n=1}^{\infty} c_n \psi_n(x), \quad c_n = \int_0^a \psi_n(x) f(x) dx.$$

The coefficients c_n carry physical meaning:

$$|c_n|^2$$

is the probability of finding the particle in the stationary state $\psi_n(x)$ with corresponding energy E_n .

4.1.7 Stationary States

The full time-dependent solutions (stationary states) are

$$\Psi_n(x, t) = \psi_n(x) e^{-iE_n t/\hbar} = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right) e^{-i\frac{n^2\pi^2\hbar}{2ma^2}t}.$$

Summary

- The infinite square well confines the particle to $0 \leq x \leq a$.
- Solutions inside the well are sinusoidal, vanishing at the boundaries.
- Allowed energies are quantized: $E_n \propto n^2$.
- Normalized eigenfunctions are standing waves of the form $\sin(n\pi x/a)$.
- The eigenfunctions are orthonormal and complete, forming a basis for expansions of arbitrary wavefunctions in the well.

Problems: One-Dimensional Infinite Square Well

Problem 1 The initial wavefunction at time $t = 0$ for a particle in a one-dimensional potential well of width a is given as,

$$\Psi(x, 0) = \begin{cases} A x(a - x), & 0 \leq x \leq a, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Find A by normalization. (b) Find the expansion coefficients c_n in $\Psi(x, 0) = \sum_n c_n \psi_n(x)$ where ψ_n correspond to the stationary states. (c) State which c_n vanish and give the closed form of the nonzero ones.

Solution. (a) Normalize:

$$1 = |A|^2 \int_0^a x^2(a - x)^2 dx = |A|^2 \frac{a^5}{30} \implies A = \sqrt{\frac{30}{a^5}}.$$

(b) The coefficients are

$$c_n = \int_0^a \psi_n(x) \Psi(x, 0) dx = \sqrt{\frac{2}{a}} A \int_0^a x(a - x) \sin\left(\frac{n\pi x}{a}\right) dx.$$

Carrying out the integral (integration by parts twice or using standard integrals) one finds (after simplification) (**please verify!!**)

$$c_n = \frac{8\sqrt{15}}{(n\pi)^3} \quad \text{for } n \text{ odd}, \quad c_n = 0 \quad \text{for } n \text{ even}.$$

(c) Thus only odd n appear; e.g. $c_1 = \frac{8\sqrt{15}}{\pi^3}$, and the probability to measure E_1 is $|c_1|^2 \approx 0.9986$ (the majority of the weight is in the ground state for this particular $f(x)$).

The time evolution follows immediately:

$$\Psi(x, t) = \sum_{n \text{ odd}} c_n \psi_n(x) e^{-iE_n t/\hbar}.$$

Problem 2 In the above problem, find the expectation values of position, energy and momentum of the particle at $t = 0$. Also, find the uncertainties in position and momentum. Verify the uncertainty relation.

Hints:

Given

$$\Psi(x, 0) = \begin{cases} A x(a - x), & 0 \leq x \leq a, \\ 0, & \text{otherwise} \end{cases}$$

Normalization gives

$$A = \sqrt{\frac{30}{a^5}}$$

For the given wave function that is non-zero only in the interval $0 \leq x \leq a$, the expectation value of any operator $\hat{O}$ is (at $t = 0$),

$$\langle \hat{O} \rangle = \int_0^a \Psi^*(x, 0) \hat{O} \Psi(x, 0) dx$$

Expectation value of position

$$\langle x \rangle = \int_0^a \Psi^*(x, 0) x \Psi(x, 0) dx$$

Expectation value of momentum

$$\langle p_x \rangle = \int_0^a \Psi^*(x, 0) \left(-i\hbar \frac{\partial}{\partial x} \right) \Psi(x, 0) dx$$

Expectation value of energy

$$\langle E \rangle = \langle H \rangle \quad \text{and use} \quad \hat{H}\Psi_n = E_n\Psi_n$$

to evaluate

$$\langle H \rangle = \int_0^a \Psi^*(x, 0) \hat{H} \Psi(x, 0) dx$$

To compute the uncertainties, first calculate

$$\langle x^2 \rangle = \int_0^a \Psi^*(x, 0) x^2 \Psi(x, 0) dx$$

$$\langle p_x^2 \rangle = \int_0^a \Psi^*(x, 0) \left(-\hbar^2 \frac{\partial^2}{\partial x^2} \right) \Psi(x, 0) dx$$

Uncertainty in position

$$\sigma_x = \sqrt{\langle x^2 \rangle - \langle x \rangle^2}$$

Uncertainty in momentum

$$\sigma_{p_x} = \sqrt{\langle p_x^2 \rangle - \langle p_x \rangle^2}$$

Finally verify the **Heisenberg uncertainty relation**

$$\sigma_x \sigma_{p_x} \geq \frac{\hbar}{2}$$

Problem 3 Consider a particle in a one-dimensional box of width a . It is given that at $t = 0$ the particle is equally likely to be anywhere in the left half:

$$\Psi(x, 0) = \begin{cases} A, & 0 \leq x \leq a/2, \\ 0, & a/2 < x \leq a. \end{cases}$$

(a) Find A . (b) Find the expansion coefficients c_n in $\Psi(x, 0) = \sum_n c_n \psi_n(x)$ where ψ_n correspond to the stationary states. (c) What is the probability to measure the ground-state energy E_1 ?

Solution. (a) Normalization:

$$1 = |A|^2 \int_0^{a/2} dx = |A|^2 \frac{a}{2} \implies A = \sqrt{\frac{2}{a}}.$$

(b) Coefficients

$$c_n = \int_0^a \psi_n(x) \Psi(x, 0) dx = \sqrt{\frac{2}{a}} A \int_0^{a/2} \sin\left(\frac{n\pi x}{a}\right) dx.$$

With $A = \sqrt{2/a}$ the prefactor becomes $2/a$, so

$$c_n = \frac{2}{a} \int_0^{a/2} \sin\left(\frac{n\pi x}{a}\right) dx = \frac{2}{a} \cdot \frac{a}{n\pi} \left(1 - \cos\frac{n\pi}{2}\right) = \frac{2}{n\pi} \left(1 - \cos\frac{n\pi}{2}\right).$$

This compact form shows the selection structure; evaluating the first few:

$$c_1 = \frac{2}{\pi}, \quad c_2 = \frac{2}{\pi}, \quad c_3 = \frac{2}{3\pi}, \quad c_4 = 0, \quad c_5 = \frac{2}{5\pi}, \quad \dots$$

(c) Probability to measure E_1 :

$$P(E_1) = |c_1|^2 = \left(\frac{2}{\pi}\right)^2 = \frac{4}{\pi^2} \approx 0.4053.$$

The full time evolution is $\Psi(x, t) = \sum_n c_n \psi_n(x) e^{-iE_n t/\hbar}$.

Remarks.

- In any expansion $\Psi(x, 0) = \sum_n c_n \psi_n(x)$, the coefficient c_n is given by $c_n = \int_0^a \psi_n^*(x) \Psi(x, 0) dx$, and $|c_n|^2$ is the probability that an energy measurement yields E_n .

Problem 4

A particle is confined in a one-dimensional infinite potential well of width L . The energy levels of the particle are given by

$$E_n = \frac{n^2\pi^2\hbar^2}{2mL^2}, \quad n = 1, 2, 3, \dots$$

Suppose the particle makes a transition from the state $n = 3$ to the state $n = 2$ by emitting a photon.

- (a) Find the energy of the emitted photon.
- (b) Determine the wavelength of the emitted photon.

Solution

The energy of the emitted photon equals the difference between the initial and final energy levels:

$$E_\gamma = E_i - E_f$$

For the transition $n = 3 \rightarrow n = 2$,

$$E_3 = \frac{9\pi^2\hbar^2}{2mL^2}, \quad E_2 = \frac{4\pi^2\hbar^2}{2mL^2}$$

Therefore,

$$E_\gamma = E_3 - E_2$$

$$E_\gamma = \frac{9\pi^2\hbar^2}{2mL^2} - \frac{4\pi^2\hbar^2}{2mL^2}$$

$$E_\gamma = \frac{5\pi^2\hbar^2}{2mL^2}$$

The energy of a photon is related to its wavelength by

$$E_\gamma = \frac{hc}{\lambda}$$

Thus,

$$\lambda = \frac{hc}{E_\gamma}$$

Substituting the expression for E_γ ,

$$\begin{aligned} \lambda &= \frac{hc}{\frac{5\pi^2\hbar^2}{2mL^2}} \\ \lambda &= \frac{2mL^2hc}{5\pi^2\hbar^2} \end{aligned}$$

Using $\hbar = \frac{h}{2\pi}$, this can also be written as

$$\lambda = \frac{8mL^2c}{5h}$$

Final Result

$$E_\gamma = \frac{5\pi^2\hbar^2}{2mL^2}, \quad \lambda = \frac{8mL^2c}{5h}$$

This is the wavelength of the photon emitted when the particle transitions from the $n = 3$ state to the $n = 2$ state in a one-dimensional infinite potential well.

4.2 Extension to the Three-Dimensional Case**4.2.1 The Potential**

In three dimensions, the infinite square well (or *box*) with cubic symmetry is defined as

$$V(x, y, z) = \begin{cases} 0, & 0 \leq x, y, z \leq a, \\ \infty, & \text{otherwise.} \end{cases}$$

Thus the particle is confined within a cube of side length a .

4.2.2 Separation of Variables

Inside the cube, the time-independent Schrödinger equation is

$$-\frac{\hbar^2}{2m} \left(\frac{\partial^2 \Psi}{\partial x^2} + \frac{\partial^2 \Psi}{\partial y^2} + \frac{\partial^2 \Psi}{\partial z^2} \right) = E\Psi.$$

We seek separable solutions of the form

$$\Psi(x, y, z) = X(x)Y(y)Z(z).$$

Substituting and dividing by $X(x)Y(y)Z(z)$ give

$$\frac{1}{X} \frac{d^2 X}{dx^2} + \frac{1}{Y} \frac{d^2 Y}{dy^2} + \frac{1}{Z} \frac{d^2 Z}{dz^2} = -\frac{2mE}{\hbar^2}.$$

Each term must be a constant:

$$\frac{d^2 X}{dx^2} = -k_x^2 X, \quad \frac{d^2 Y}{dy^2} = -k_y^2 Y, \quad \frac{d^2 Z}{dz^2} = -k_z^2 Z,$$

with

$$E = \frac{\hbar^2}{2m} (k_x^2 + k_y^2 + k_z^2).$$

4.2.3 Boundary Conditions and Solutions

The boundary conditions are

$$X(0) = X(a) = 0, \quad Y(0) = Y(a) = 0, \quad Z(0) = Z(a) = 0.$$

Thus the solutions are

$$X_{n_x}(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n_x \pi x}{a}\right), \quad Y_{n_y}(y) = \sqrt{\frac{2}{a}} \sin\left(\frac{n_y \pi y}{a}\right), \quad Z_{n_z}(z) = \sqrt{\frac{2}{a}} \sin\left(\frac{n_z \pi z}{a}\right),$$

where $n_x, n_y, n_z = 1, 2, 3, \dots$

4.2.4 Wavefunctions and Energies

The normalized eigenfunctions of the 3D cube are

$$\Psi_{n_x, n_y, n_z}(x, y, z) = \left(\frac{2}{a}\right)^{3/2} \sin\left(\frac{n_x \pi x}{a}\right) \sin\left(\frac{n_y \pi y}{a}\right) \sin\left(\frac{n_z \pi z}{a}\right),$$

with corresponding energies

$$E_{n_x, n_y, n_z} = \frac{\hbar^2 \pi^2}{2ma^2} (n_x^2 + n_y^2 + n_z^2).$$

4.2.5 Degeneracy

Because the energy depends only on the sum of squares $n_x^2 + n_y^2 + n_z^2$, different triplets (n_x, n_y, n_z) can yield the same energy. For example,

$$(n_x, n_y, n_z) = (1, 1, 2), \quad (1, 2, 1), \quad (2, 1, 1)$$

all correspond to the same energy level. This threefold degeneracy is a direct consequence of the cubic symmetry of the potential.

4.3 Quantum Tunneling

Quantum tunnelling through a rectangular barrier

In classical mechanics, if a particle of energy E encounters a potential barrier of height $U > E$, it is completely reflected; the barrier is impenetrable. However, in quantum mechanics, the particle is described by a wavefunction that extends into the classically forbidden region. As a result, there exists a finite probability for the particle to penetrate the barrier and emerge on the other side - a purely quantum effect known as **tunnelling**.

Potential barrier and setup

Consider a particle of mass m and energy E incident from the left on a one-dimensional rectangular potential barrier of height U and width L :

$$V(x) = \begin{cases} 0, & x < 0 \quad (\text{Region I}) \\ U, & 0 \leq x \leq L \quad (\text{Region II}) \\ 0, & x > L \quad (\text{Region III}) \end{cases}$$

We assume $E < U$, i.e., the particle energy is less than the potential height.

Time-independent Schrödinger equation. In each region, the stationary wavefunction $\psi(x)$ satisfies

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + V(x)\psi = E\psi.$$

This can be written as

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2} (E - V(x))\psi = 0.$$

Solutions in different regions**Region I** ($x < 0$, $V = 0$):

$$\frac{d^2\psi}{dx^2} + k_1^2\psi = 0, \quad k_1 = \frac{\sqrt{2mE}}{\hbar}.$$

The general solution is a superposition of incident and reflected waves:

$$\psi_I(x) = e^{ik_1x} + r e^{-ik_1x}.$$

Here r is the reflection amplitude.

Region II ($0 \leq x \leq L$, $V = U$):

$$\frac{d^2\psi}{dx^2} - k_2^2\psi = 0, \quad k_2 = \frac{\sqrt{2m(U - E)}}{\hbar}.$$

Since $E < U$, the term $(E - U)$ is negative, making k_2 real and positive. Hence the solution is no longer oscillatory, but exponential:

$$\psi_{II}(x) = A e^{k_2x} + B e^{-k_2x}.$$

This describes a wave that decays (and grows) exponentially within the barrier. The decaying part represents the *evanescent wave* - a nonzero amplitude within the classically forbidden region.

Region III ($x > L$, $V = 0$):

$$\frac{d^2\psi}{dx^2} + k_1^2\psi = 0,$$

giving a transmitted wave:

$$\psi_{III}(x) = t e^{ik_1x},$$

where t is the transmission amplitude.

Boundary conditions

At $x = 0$ and $x = L$, both ψ and $\frac{d\psi}{dx}$ must be continuous:

$$\begin{aligned} \psi_I(0) &= \psi_{II}(0), & \psi'_I(0) &= \psi'_{II}(0), \\ \psi_{II}(L) &= \psi_{III}(L), & \psi'_{II}(L) &= \psi'_{III}(L). \end{aligned}$$

Solving these four equations for r , t , A , B yields (after some algebra) the transmission probability:

$$T = |t|^2 = \left[1 + \frac{U^2 \sinh^2(k_2L)}{4E(U - E)} \right]^{-1}.$$

This is the exact quantum-mechanical transmission coefficient for a rectangular barrier.

Opaque-barrier approximation

If the barrier is *thick* or *high* so that $k_2L \gg 1$, then

$$\sinh(k_2L) \approx \frac{1}{2}e^{k_2L}.$$

Substituting in the exact formula gives

$$T \approx \left(\frac{4\sqrt{E(U-E)}}{U} \right)^2 e^{-2k_2L}.$$

For qualitative estimates, the prefactor is often neglected, yielding the approximate exponential dependence:

$$T \propto e^{-2k_2L}, \quad k_2 = \frac{\sqrt{2m(U-E)}}{\hbar}.$$

Thus, the transmission probability falls off exponentially with both the *width* and the *height* of the potential barrier.

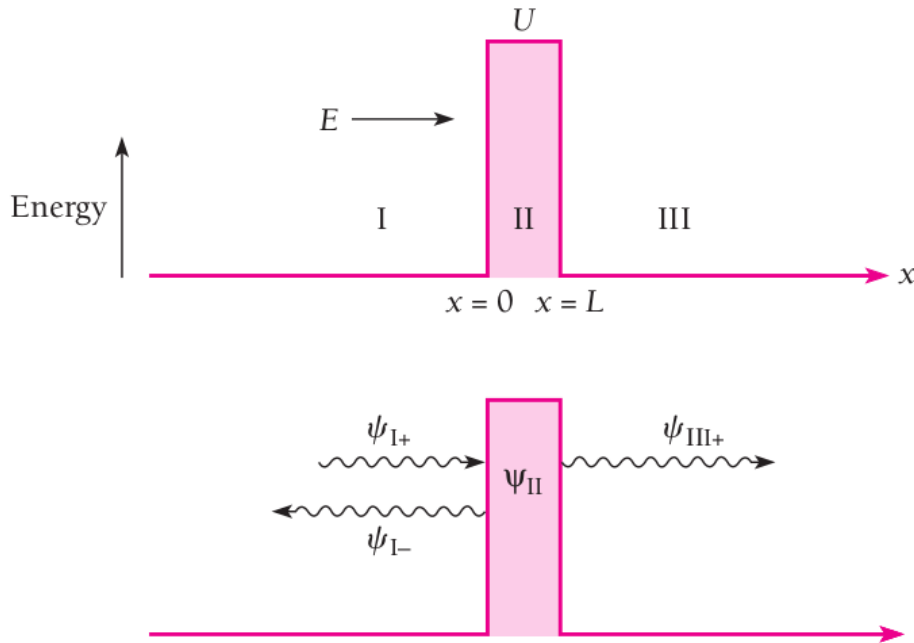


Figure 4.3: Rectangular barrier of height U and width L . A particle of energy $E < U$ has a finite probability of tunnelling through the barrier. (Adapted from *Modern Physics* by Arthur Beiser, 6th edition, for instructional use only.)

Example: Scanning Tunneling Microscope (STM). The STM operates on the principle of quantum tunneling. When a sharp metallic tip is brought extremely close (a few Angstroms) to a conducting surface, electrons can tunnel through the vacuum gap between them, producing a measurable tunneling current.

The tunneling current I depends sensitively on the separation L as

$$I \propto e^{-2k_2L}, \quad k_2 = \frac{\sqrt{2m\phi}}{\hbar},$$

where ϕ is the effective work function (barrier height).

Because of this exponential dependence, a change in distance of even 0.1 nm can change the current by an order of magnitude, allowing the STM to resolve surface features with atomic precision.

Summary

- Quantum tunnelling arises from the wave nature of matter and the continuity of the wavefunction.
- The probability of transmission through a barrier of height U and width L is

$$T = \left[1 + \frac{U^2 \sinh^2(k_2 L)}{4E(U - E)} \right]^{-1}.$$

- For $k_2 L \gg 1$, $T \approx e^{-2k_2 L}$.
- Applications include α -decay, STM, tunnel diodes, and field emission.